FINANCE I

Facultad de Economía y Negocios
Universidad de Chile

Prof.: Alejandro Bernales

Exam

Time Allowed: 1.5 Hours

Instructions:

The Exam has 3 questions, each of them have the same weight.

PLEASE WRITE YOUR ID NUMBER OR PASSPORT IN ALL THE PAGES.

PLEASE ONLY USE THE SPACE IN EACH QUESTION TO ANSWER ALL SECTIONS (USE ONLY ONE SIDE OF THE PAGE - ONE LANGUAGE)

The space provided is sufficient to give a concise and pertinent answer to each question and/or sub-point of a question.

Extra answer booklets will not be provided – please use the spaces provided only. Answers written outside the spaces or on other booklets will not be marked.

Electronic calculators may be used provided that they cannot store text

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Time Allowed: 1.5 Hours

Instrucciones:

El Examen tiene 3 preguntas, cada una de ellas tienen la misma ponderación.

POR FAVOR ESCRIBA SU RUT O PASAPORTE EN TODAS LAS HOJAS.

POR FAVOR UTILICE EL ESPACIO PROVISTO EN CADA PREGUNTA PARA RESPONDER CADA SECCIÓN (USAR UN SOLO LADO DE LA HOJA - UN SOLO LENGUAJE).

El espacio entregado es suficiente para entregar una respuesta concisa y pertinente a cada pregunta y/o sub-pregunta.

No se entregarán hojas adicionales para responder por favor utilice solamente el espacio entregado. Respuestas escritas fuera de este espacio o en otras hojas no serán revisadas.

Calculadoras electrónicas pueden ser utilizadas salvo las que puedan almacenar texto

Question I. A stock price is currently \$30. During each 2-month period it will increase by 8% or reduce by 10% (for the next 4 months). The risk-free interest rate is 5%. Use a two-step tree to calculate the value of a derivative that pays off in 4 months:

$$Payoff = [\max(30 - S_T, 0)]^2,$$

where S_T is the stock price in 4 months (i.e. an European-Style derivative). If the derivative is American-style, should it be exercised early?

Answer

This type of option is known as a power option. A tree describing the behavior of the stock price is shown in Figure M . The risk-neutral probability of an up move, p, is given by

$$p = \frac{e^{0.05\times 2/12} - 0.9}{1.08 - 0.9} = 0.6020$$

Calculating the expected payoff and discounting, we obtain the value of the option as

$$[0.7056 \times 2 \times 0.6020 \times 0.3980 + 32.49 \times 0.3980^2]e^{-0.05\times 4/12} = 5.394$$

The value of the European option is 5.394. This can also be calculated by working back through the tree as shown in Figure M . The second number at each node is the value of the European option.

Early exercise at node C would give 9.0 which is less than 13.2449. The option should therefore not be exercised early if it is American.

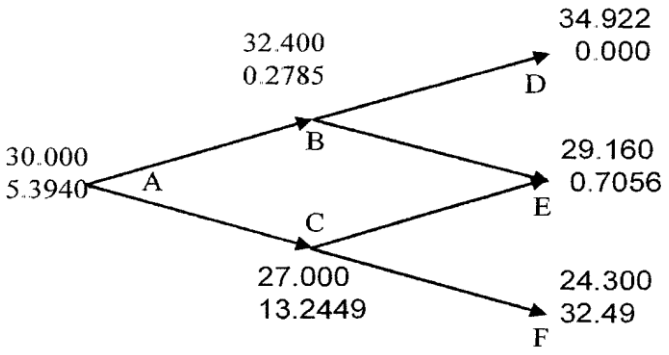


Figure M Tree to evaluate European power option
At each node, upper number is the stock price; next number is the option price

Pregunta I. Una acción actualmente vale $30, la cual cada 2 meses subirá un 8% o decrecerá un 10% (lo que ocurrirá por los próximos 4 meses). La tasa libre de riesgo es de un 5% anual. Usando un árbol binomial de dos pasos, calcule el valor (al día de hoy) de un derivado que paga ("payoff") a la madurez en 4 meses:

$$Payoff = [\max(30 - S_T, 0)]^2,$$

donde S_T es el valor de la acción en 4 meses más (i.e. un derivado del estilo "Europeo"). ¿Si el derivado fuera del tipo de ejercicio "Americano", debiese ser ejercido anticipadamente?

Respuesta

Question II.

2a) Option prices are calculated using a risk neutral valuation, by assuming that all agents in the economy have a null risk aversion. For that reason, it is possible to directly discount options' payoffs at the risk-free rate. Please, explain potential problems of assuming a risk neutral valuation in which all agents do not have risk aversion.

Answer

This sentence is wrong. The risk neutral valuation does not assume that all agents in the economy have a null risk aversion. In fact, agents can have different levels of risk aversion.

The risk neutral valuation is a technique, in which the risk is eliminated by using a portfolio. Thus, the concept 'risk neutral valuation' is related to eliminate risk rather than assuming a null level of risk aversion in the economy. This risk neutral portfolio has the option contract and a quantity of stock shares. After that, the payoffs are discounted to the present at the risk free rate since this portfolio does not have risk. Then, it is possible to obtain the price of the option by decomposing the portfolio in its different components.

2b) Why should students in Chile learn 'options' despite there are no developed option markets in Chile? Provide one reason.

Answer

-All insurances are options, and there are a plenty of insurance companies in Chile (e.g., Isapres and insurance companies related to car insurances, travel insurances, etc.). Thus, students in Chile in this area will face with option modelling and option pricing.

or

-Options can be replicated (in a synthetic way) by using a bond and stock; thus it is not necessary to have an option market to use them.

Pregunta II.

2a) Los precios de las opciones se calculan utilizando una valoración neutral al riesgo en la cual se asume que todos los agentes de la economía tienen una aversión nula al riesgo. Por esa razón, es posible descontar las ganancias de las opciones directamente a la tasa libre de riesgo. Por favor, explique los posibles problemas de asumir una valoración neutral al riesgo en la que todos los agentes no tienen ninguna aversión al riesgo.

Respuesta

2b) ¿Por qué los estudiantes en Chile deberían aprender 'opciones' a pesar de que no hay mercados de opciones desarrollados en Chile? Entregue una razón.

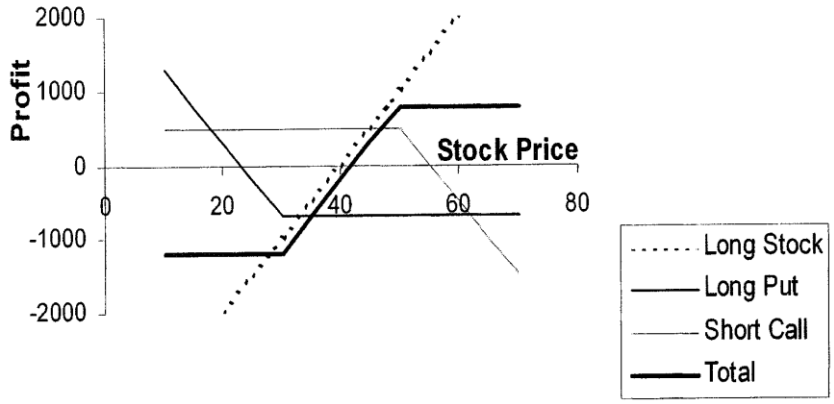
Responder

Question III.

3a) The price of a stock is $40. The price of a 1-year European put option on the stock with a strike price of $30 is quoted at $7 and the price of a 1-year European call option on the stock with a strike price of $50 is quoted at $5. Suppose that an investor buys 100 shares, shorts 1 call option contract, and buys 1 put option contract. Draw a diagram illustrating how the investor's profit or loss varies with the stock price over the next year.

Hint: remember that when you buy (or sell) option contracts, they always involve #100 underlying assets (i.e. in this case #100 shares). Thus, in the example above, the value of 1 call option contract with price c is in reality $c * 100$ (because, it involves 100 underlying assets).

Answer:



The figure above shows the way in which the investor's profit varies with the stock price in the first case. For stock prices less than $30 there is a loss of $1200 (i.e. 500(selling call contract)-700(buying put contract)-10*100(losses from the stock)). As the stock price increases from $30 to $50 the profit increases from -1200 to $800. Above $50 the profit is $800.

Pregunta III.

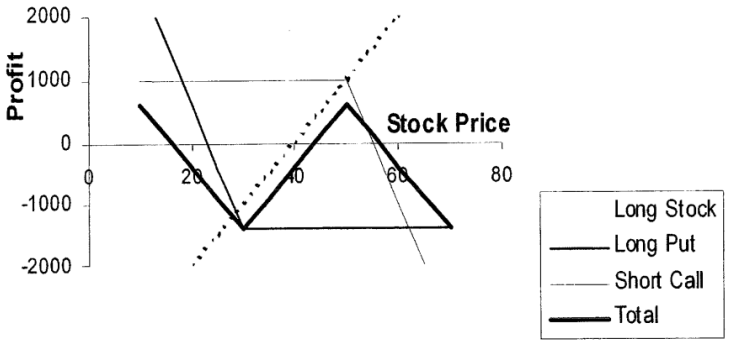
3a) El precio de una acción es de $40. El precio de una opción Put Europea sobre la acción descrita, con madurez a 1 año y un precio Strike de $30, es de $7; mientras que el precio de una opción Call Europea sobre la misma acción, con madurez a 1 año y un precio Strike de $50, es de $5. Suponga que un inversionista compra 100 acciones, vende 1 contrato Call, y compra 1 contrato Put. Dibuje un diagrama indicando cómo las ganancias o pérdidas del inversionista varían con el precio de la acción durante el siguiente año.

Hint: recuerde que cuando compra (o vende) contratos de opciones, siempre involucran #100 activos subyacentes (i.e. #100 acciones en este caso). Así, en el ejemplo anterior, el valor de 1 contrato de opción de compra con precio c es en realidad $c * 100$ (porque involucra 100 activos subyacentes).

Respuesta:

3b) How does your answer change if instead the investor buys 100 shares, shorts 2 call option contracts, and buys 2 put option contracts?

Answer



In the second case pattern has a zigzag shape, because calls and puts involve each other 200 underlying assets, while the investor only has 100 shares. Thus, for example, when the price is below 30, the line going below (from the put) and the line going up (from the stock) are not eliminated.

3b) ¿Cómo cambia su respuesta si el inversionista, en vez de comprar el portafolio anterior, compra 100 acciones, vende 2 contratos Call, y compra 2 contratos Put?

Respuesta